

James (Huck) Rees

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BACKGROUND

PhD candidate in Geography with extensive expertise in geomorphology, hydrology, and remote sensing, specializing in modelling sediment storage and transport, surface water detection, mapping river corridor and floodplain evolution, and quantifying sediment and carbon storage in beaver-mediated river systems. Skilled in Python, R, and geospatial programming, with a proven record of interdisciplinary collaboration, grant success, and peer-reviewed publications. Passionate about applying advanced geospatial analytics and field-validated insights to create practical, scalable tools that support wetland restoration, flood protection, and carbon storage.

SKILLS

Programming languages: Python (geopandas, shapely, rasterio, geemap, xarray, scikit-learn, whitebox, gdal, scipy); R (lidR), JavaScript

Software/toolkits: Claude Code, Google Earth Engine Python API, ArcGIS Pro, QGIS, GitHub, HEC-RAS, HEC-LifeSim, HEC-FIA, Hazus, ENVI, Microsoft Office Suite, Affinity Designer, Adobe

Sensors and datasets: Landsat, Sentinel 1/2, PlanetScope, NAIP, EMIT, SWOT, 3DEP, various LiDAR/DEMs, HydroSHEDS suite, SWORD, NASADEM, NLCD, Dynamic Surface Water Extent (DSWE)

EDUCATION

In progress	Ph.D. in Geography , University of California, Santa Barbara Dissertation: <i>Calibrating the geomorphic clock: a remote sensing-based approach to modelling sediment storage and routing in alluvial river corridors</i> GPA: 4.0
2023	M.A. in Fluvial Geomorphology , University of Colorado, Boulder Thesis: <i>Sediment and Carbon Storage and Aggradation Rates within Beaver-mediated River Corridors in the Southern Rocky Mountains, USA</i> Certificate in Hydrologic Sciences GPA: 4.0
2019	B.A. in Geology <i>summa cum laude</i> , University of California, Davis Phi Beta Kappa Thesis: <i>Managed retreat and community relocation as a flood mitigation measure in small Midwest communities</i> B.A. in History <i>cum laude</i> Minor in Geographic Information Science GPA: 3.84

PROFESSIONAL EXPERIENCE

Sep. 2023 to Present	Graduate Researcher , Water, Vegetation, and Society Lab/Surface Processes Lab University of California, Santa Barbara – Department of Geography
June 2026 to Aug. 2026	Summer Intern , Imaging Spectroscopy Group NASA Jet Propulsion Laboratory
July 2021 to Aug. 2023	Graduate Researcher , Fluvial Ecogeomorphology Lab University of Colorado, Boulder – Department of Geography
Aug. 2019 to July 2021	Water Resources Coordinator , Water Specialty Group HDR Engineering
June 2019 to Aug. 2019	Research Assistant , Childs Meadow Restoration Project University of California, Davis – Center for Watershed Science
Oct. 2017 to June 2019	Undergraduate Researcher , Geomorphology and Hazard Research Group University of California, Davis – Department of Earth and Planetary Sciences
May 2017 to Oct 2018	Research Assistant , Map Collection Room University of California – Shields Library

HONORS AND AWARDS

2025	Top Cited Author Award, River Research and Applications
2023	Sustainability Award, CU Boulder
2021	NSF Graduate Research Fellowship Honorable Mention
2019	Highest Honors Distinction, UC Davis Department of Earth and Planetary Sciences
2019	Honors Distinction, UC Davis History Department
2019	Outstanding Senior, UC Davis Department of Earth and Planetary Sciences
2019	Winner, Prized Writing Competition, Scientific and Technical Writing
2019	Department Citation, UC Davis History Department
2018	Phi Beta Kappa Honors Society

GRANTS

2025	Benioff Ocean Sciences Laboratory Research Grant (\$6,000)
2024	Schmidt Family Foundation Research Accelerator Fellowship (\$8,000)
2022	Graduate Research Fellowship, Rocky Mountain Biological Lab (\$500)
2022	Adam Kolff Memorial Research Scholarship, CU Boulder (\$5,000)
2022	Jeanne DeSana Graduate Research Scholarship, CU Boulder (\$5,000)
2020	Fulbright Open Study/Research Grant, South Pacific Island Nations (~\$20,000)
2019	Research Travel Grant, to present at GSA, UC Davis (\$500)
2018	Everett Dale Jackson Memorial Scholarship, UC Davis (\$5,000)
2018	Michael Alan Anthony Grant, UC Davis (\$5,000)

PUBLICATIONS

Rees, H., Crompton, O., Lourenço, M., WinklerPrins, L., Larsen, L., Caylor, K. K., Ganti, V., submitted. *A 41-year monthly inundation record reveals abrupt redistribution of flood waters across the Okavango Delta since 2012*. In review at Nature Communications.

Rees, H., Ganti, V., Greenberg, E., Li, G., Caylor, K., submitted. *Calibrating the geomorphic clock: Remote sensing of sediment storage and transit timescales in alluvial river corridors*. In review at Geophysical Research Letters.

McLeod, J.S., Ganti, V., Hampson, G.J., Bell, R.E., Slater, L.J., **Rees, H.**, Whittaker, A.C., submitted. *Global sediment transport intermittency is set by river planform*. In review at Nature Geoscience

Rees, H., Thompson, D.R., Eckert, R., Greenberg, E.B., & Green, R.O. *in prep. Radiometric stability of the EMIT imaging spectrometer over North African pseudo-invariant calibration sites*. To be submitted to Remote Sensing of Environment.

Rahman, S., Chamberlain, L., ... **Rees, H.**, ... Zhang, L., *in prep Leaky deltas: sources or sinks in the global carbon cycle?* To be submitted to Global Biogeochemical Cycles

WinklerPrins, L., Fonetenot, K... **Rees, H.**, et al., *in prep. Early Career Perspectives on the next generation of deltaic science*. To be submitted to Frontiers in Marine Science

WinklerPrins, L., Yang, W., Crompton, O., Lourenço, M., Ruer, M., Taylor, R., **Rees, H.**, Larsen, L., in prep. *A Validated Land Cover Classification of the Okavango Delta, 2023*. To be submitted to Remote Sensing of Environment.

Rees, J., Lininger, K.B., Landis, J., Briles, C., 2024. *Assessing controls on sedimentation rates and sediment organic carbon accretion in beaver ponds*. Science of the Total Environment.

Ockelford, A., Wohl, E., ... **Rees, J.**, ... Aarnink, J., 2024. *Working with wood in rivers in the Western United States*. River Research and Applications.

Pinter, N., **Rees, J.**, 2021. *Measuring success for managed flood retreat and community relocation in the Midwest USA*. Natural Hazards.

Pinter, N., Ishiwatari, M., Nonoguchi, A., Tanaka, Y., Casagrande, D., Durden, S., **Rees, J.**, 2019. *Large-scale managed retreat and structural protection following the 2011 Japan tsunami*. Natural Hazards.

Rees, J., 2019. *Literature review of mechanisms and timescales involved in the termination of the African Humid Period*. Prized Writing, University of California, Davis.

PRESENTATIONS AND TALKS

Rees, H., Ganti, V., Greenberg, E., Li, G., Caylor, K., (December 2026) Channel slope and planform set the storage duration of floodplain sediment in alluvial river corridors. To be presented at the American Geophysical Union fall Meeting. San Francisco, CA.

Rees, H., Lourenço, M., WinklerPrins, L., Larsen, L., Caylor, K. K., Ganti, V., (December 2026) A 41-year satellite record reveals abrupt redistribution of the Okavango Delta's flood pulse since 2012. To be presented at the American Geophysical Union fall Meeting. San Francisco, CA.

Thompson, D.R., Eckert, R., **Rees, J.**, Brodrick, P., et al. (September 2026). EMIT instrument and calibration. Presented at the EMIT Science and Applications Technical Interchange Meeting. Pasadena, CA.

Hampton, J., Evans, K., ... **Rees, H.**, ... et al., (April 2026) Exploring International Freshwater Ecosystem Management Strategies for New Perspectives: the Noce River, Italy and Yuba River, California, USA. Presented at the European Geophysical Union annual Meeting. Vienna, Austria.

Rees, H., Ganti, V., Greenberg, E., Li, G., Caylor, K., (April 2026) Planform and channel migration rate drive floodplain sediment storage durations and transit velocities. Presented at the Southern California Geomorphology Symposium. Riverside, CA.

Rees, H., Ganti, V., Greenberg, E., Li, G., Caylor, K., (December 2025) Calibrating the geomorphic clock: Remote sensing of sediment storage and transit timescales in alluvial river corridors. Presented at the American Geophysical Union fall Meeting. New Orleans, LA.

Triantafyllou, S., **Rees, H.**, et al., (December 2025) International Approaches to Freshwater Ecosystems: the Noce River, Italy and Yuba River, California, USA. Presented at the American Geophysical Union fall Meeting. New Orleans, LA.

Scamardo, J., Lininger, K.B., **Rees, H.**, (December 2025) Synthesizing Drivers of Beaver Dam Sediment Storage Across Headwater Systems. Presented at the American Geophysical Union fall Meeting. New Orleans, LA.

Scamardo, J., Lininger, K.B., **Rees, H.**, (October 2025) How much sediment can a beaver dam store: synthesizing drivers of beaver pond sedimentation and predicting sediment storage benefits of beaver dams. Presented at the 7th International Symposium on River Science. Davis, CA.

Rees, H., Ganti, V., Greenberg, E., Li, G., Caylor, K., (April 2025) Calibrating the geomorphic clock: Remote sensing of sediment storage and transit timescales in alluvial river corridors. Presented at the Southern California Geomorphology Symposium. Los Angeles, CA.

Rees, J., Ganti, V., and Caylor, K., (December 2024). Natural and anthropogenic controls on floodplain reworking timescales and transient sediment storage in lowland river corridors. Presented at the American Geophysical Union Fall Meeting. Washington D.C.

Lininger, K.B, and **Rees, J.**, (October 2024). Sedimentation & Carbon Accretion in Beaver Ponds & Implications for Restoration. Presented at BeaverCon. Boulder, CO.

Rees, J., Lininger, K.B., Briles., C, Landis, J.D., (March 2023). Beaver dams as sites of carbon accretion and sediment storage and implications for river restoration. Presented at the Association of American Geographers Annual Meeting. Denver, CO.

Rees, J., Lininger, K.B., Briles., C, Landis, J.D., (December 2022). Quantifying Sediment and Carbon Storage and Aggradation Rates within Beaver-mediated River Corridors in the Southern Rocky Mountains, USA. Presented at the American Geophysical Union Fall Meeting. Chicago, IL.

Rees, J. (April 2022). Assessing managed flood retreat and community relocation as a flood mitigation and climate adaptation measure in small communities. Presented at the CU Annual Hydrologic Sciences Symposium. Boulder, CO.

Ishiwatari, M., Pinter, N., Casagrande, D., and **Rees, J.** (August 2021). Relocation programs; making resilient communities through involvement of diversified groups in Japan and the US. Presented at the 8th International Conference on Flood Management. Iowa City, IA.

Rees, J. and Pinter, N. (May 2019). The effectiveness of managed retreat and community relocation as a flood mitigation measure in small Midwest communities. Presented at the GSA Cordilleran Section Annual Meeting. Portland, OR.

Rees, J. (February 2019). Field Data Collection with QField. Presented at #MaptimeDavis at the UC Davis Center for Spatial Sciences. Davis, CA.

MEDIA COVERAGE

Cheshire, C., 2022. *As CU Hosts Global Climate Summit, Students Call for Fossil Fuel Divestment*. Westword. <https://www.westword.com/news/university-colorado-divestment-fossil-fuels-climate-change-15592395>

Gies, E., 2022. *Water Always Wins – Thriving in an Age of Drought and Deluge*. The University of Chicago Press. <https://press.uchicago.edu/ucp/books/book/chicago/W/bo161589950.html>

Wise, C., 2019. *The radical approach these communities have taken to flood mitigation*. 2019. PBS NewsHour. <https://www.pbs.org/newshour/show/the-radical-approach-these-communities-have-taken-to-flood-mitigation>

Kerlin, K., Proudman, J., 2019. *Small Towns, Big Flood Waters*. UC Davis. <https://climatechange.ucdavis.edu/climate/news/small-towns-big-flood-waters>

Hersher, R., 2018. *Wisconsin Reservation Offers A Climate Success Story And A Warning*. NPR. <https://www.npr.org/2018/08/15/632335735/wisconsin-reservation-offers-a-climate-success-story-and-a-warning>

TEACHING EXPERIENCE

Landscapes and Water (Lead Teaching Assistant), 2 semesters

Introduction to Geographic Information Systems (Teaching Assistant), 2 quarters

Global Warming (Teaching Assistant), 1 quarter

Quantitative Geomorphology (Lead Teaching Assistant), 2 quarters

FIELD EXPERIENCE

2025	Sierra Nevada, California (3 weeks) <i>Geomorphic/hydrologic surveys and sediment coring in beaver meadows</i>
2025	Dolomites, Italy (2 weeks) <i>International approaches to freshwater management</i>
2023	Antioquia, Colombia, (2 weeks) <i>Field training and hydrologic surveys on the Magdalena River</i>
2022	Rocky Mountains, Colorado, (6 weeks) <i>Geomorphic/hydrologic surveys and sediment coring in natural beaver meadows</i>
2019	Cascade Mountains, California, (6 weeks) <i>Aerial (drone), geomorphic/hydrologic surveys to monitor beaver meadow restoration</i>
2019	Midwest USA, (2 weeks) <i>Structure surveys and stakeholder interviews for managed retreat research</i>
2018	Northern Japan, (1 week) <i>Coastal flood mitigation measures following the 2011 Tohoku Tsunami</i>

VOLUNTEER/SERVICE

2026 Course Designer, Instructor, School for Scientific Thought

2025 Reviewer, Communications Earth and Environment

2023-2025	Mentor, GEMSS Graduate Mentoring Program
2023	Reviewer, Water Resources Research
2022-2023	Cofounder, Fossil Free CU Divestment Campaign
2022-2023	Mentor, CU Boulder Graduate Mentoring Program
2022	Course Designer, Instructor, Earth Explorers Science Mentoring Program
2021	Volunteer, 350 Sacramento
2016	Volunteer, California State Senate Campaign

MEMBERSHIP

American Geophysical Union
 Association of American Geographers
 American Water Resources Association
 Phi Beta Kappa Honors Society